

Proofs should be written neatly and concisely. You may quote theorems from lecture, but any other results need to be proven. Make sure to reference all sources of collaboration.

- (1) Show that local compactness is invariant under continuous open mappings.
- (2) Let $\{X_\alpha\}$ be an indexed family of non-empty sets. Show that $\prod X_\alpha$ is locally compact if and only if each X_α is locally compact and X_α is compact for all but finitely many values of α . (You may assume the Tychonoff Theorem.)
- (3) Let $\{(X_i, \tau_i)\}_{i \in I}$ be a family of topological spaces and define the box topology on $\prod_{i \in I} X_i$ as the topology generated by the set

$$\left\{ \prod_{i \in I} U_i \mid U_i \in \tau_i \right\}.$$

- (a) Show that the product of compact topological spaces, endowed with the box topology, is not necessarily compact.
- (b) Let X be a topological space. We endow $X^{\mathbb{N}} = \prod_{i \in \mathbb{N}} X$ with the box topology. Is the diagonal map

$$\begin{aligned} X &\rightarrow X^{\mathbb{N}} \\ x &\mapsto (x, x, x, \dots) \end{aligned}$$

continuous?

- (4) Show that the one-point compactification of $\mathbb{Z}_+$ is homeomorphic to the subspace $\{0\} \cup \{\frac{1}{n} \mid n \in \mathbb{Z}_+\} \subset \mathbb{R}$.
- (5) Let X denote the rational points of the interval $[0, 1] \times \{0\} \subset \mathbb{R}^2$. Let T denote the union of all line segments joining the point $p = (0, 1)$ to the points of X .
 - (a) Show that T is path connected, but it is locally connected only at the point p .
 - (b) Find a subset of $\mathbb{R}^2$ that is path connected but is locally connected at none of its points.
- (6) Define the cone CX to be the quotient space obtained by identifying $X \times \{1\}$ in $X \times [0, 1]$ to a point. Prove that CX is path-connected. Deduce that any X is a subspace of a path connected space.